

COS30018 – Intelligent Systems

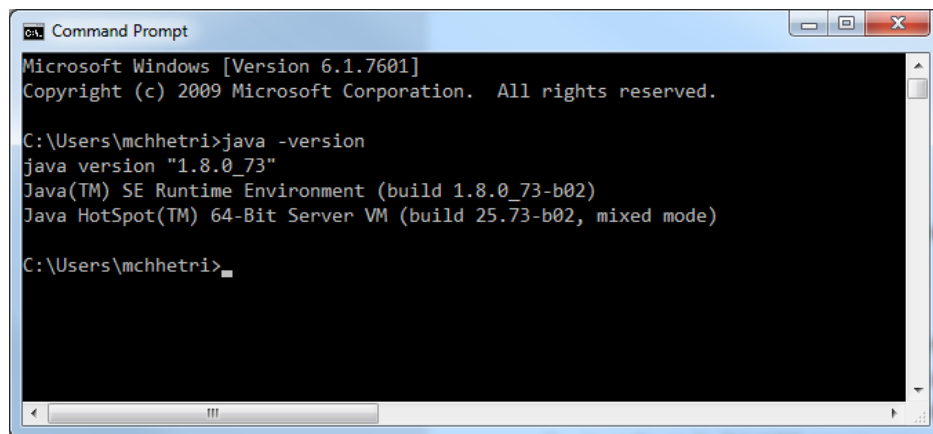
This document outlines the steps involved in setting up your machine for running the JADE middleware and for coding JADE agents. It assumes that the machine does not have Java installed already.

Install Java

In order to run JADE agents, you need to have a **Java Runtime Environment (JRE)** installed on your machine. In order to build and compile JADE agents, you need to have the **Java Development Kit (JDK)** installed as well. You can download this from the following website (<http://www.oracle.com/technetwork/java/javase/downloads/index.html>) (latest version is JDK 8). Download the right version (32 bit or 64 bit) depending upon your underlying OS.

Check your Java installation:

Once you have downloaded and installed the JDK, you can check to see if it was correctly installed from the command line as shown below



```
Microsoft Windows [Version 6.1.7601]
Copyright (c) 2009 Microsoft Corporation. All rights reserved.

C:\Users\mchhetri>java -version
java version "1.8.0_73"
Java(TM) SE Runtime Environment (build 1.8.0_73-b02)
Java HotSpot(TM) 64-Bit Server VM (build 25.73-b02, mixed mode)

C:\Users\mchhetri>
```

Figure 1 Checking Java Version

Download & Install Eclipse

The next step is to download Eclipse (<http://www.eclipse.org/downloads/packages/>). Choose the right release and package. The latest release is Oxygen and you can download either the Eclipse IDE for Java EE Developers or Eclipse IDE for Java Developers (http://www.eclipse.org/downloads/download.php?file=/technology/epp/downloads/release/oxygen/R/eclipse-java-oxygen-R-win32-x86_64.zip) . (Download the right version depending upon the underlying OS).

Extract the downloaded zipped folder to the location of your choice on your machine. In the figure below, it has been extracted to C:\Tools\eclipse-java-oxygen-R-win32-x86_64. Click on the **eclipse.exe** executable to start up the IDE.

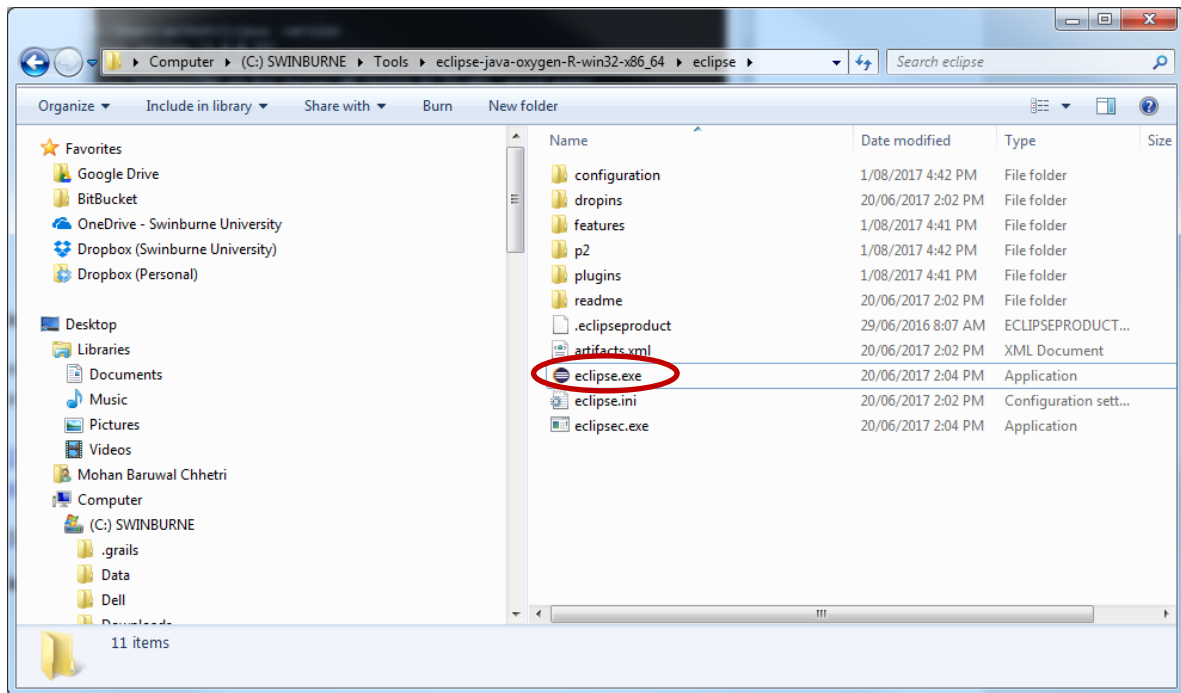


Figure 2 Extracted Eclipse Files

Download JADE

Download JADE from the website (<http://jade.tilab.com>) – current version is JADE 4.5.0 (08/06/2017). Choose the *JadeAll* file, which contains the binaries (compiled code), the source, the documentation and the examples. Then extract it (**JADE-all-4.5.0.zip**) to the location of choice on your machine. This extracted folder has 4 zipped folders.

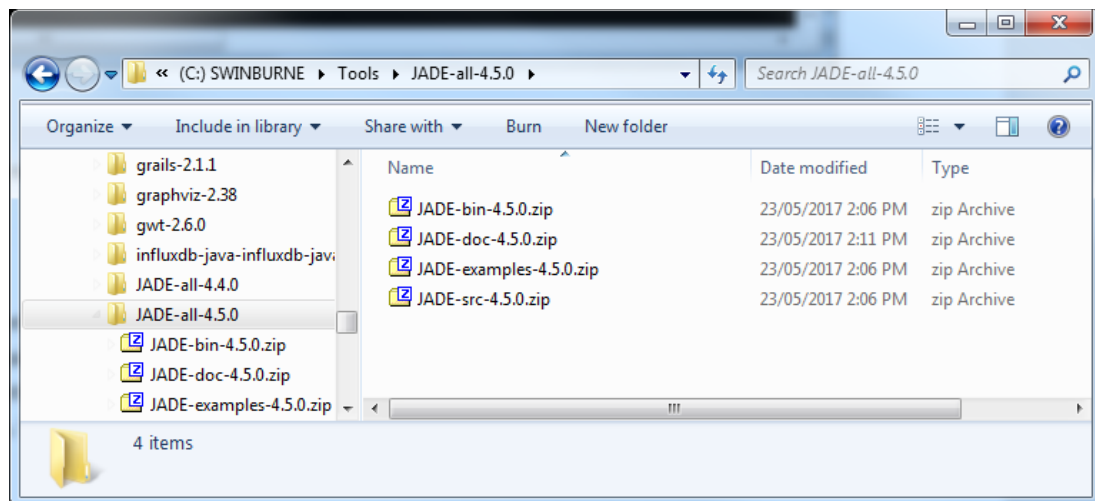


Figure 3 Downloading JADE to local machine

We are interested in **JADE-bin-4.5.0.zip**. Extract that as well.

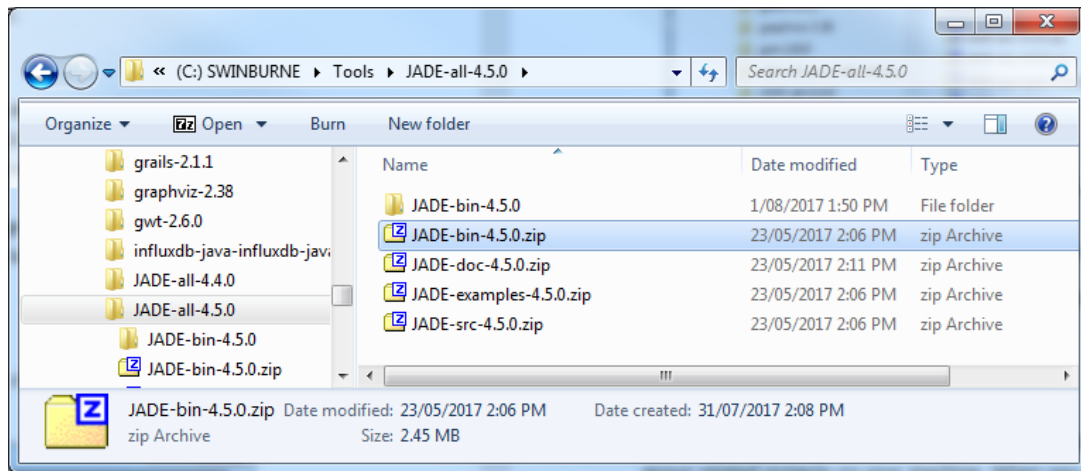


Figure 4 Extracting Binaries

Inside the extracted folder, there is a single java archive (jar) file called **jade.jar**. The entire binaries for running the JADE middleware are contained inside this single jar file.

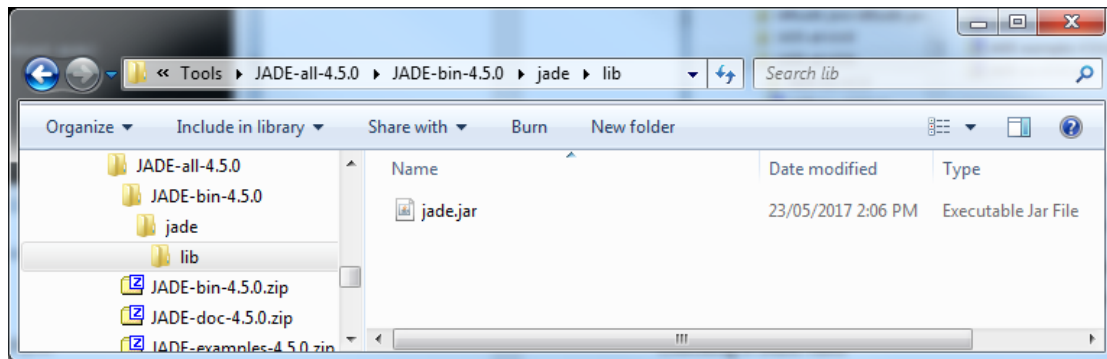
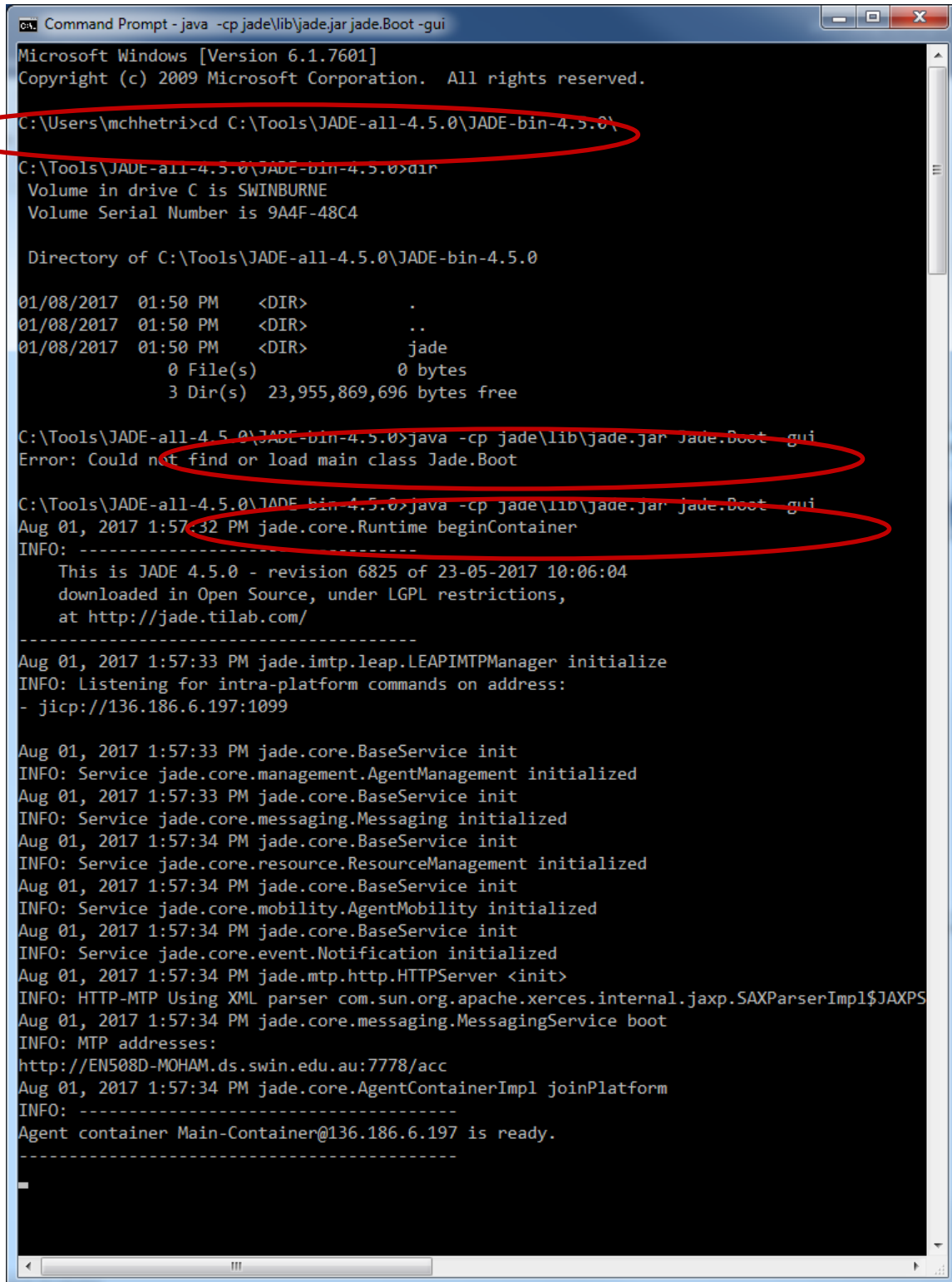


Figure 5 Single JADE binary file

Run JADE from command line

In order to check if JADE runs successfully on your machine, **open** up the command line, **change directory** to the location where JADE is installed and start up JADE as shown below. The JADE Remote Management GUI is also started up.



```
Command Prompt - java -cp jade\lib\jade.jar jade.Boot -gui
Microsoft Windows [Version 6.1.7601]
Copyright (c) 2009 Microsoft Corporation. All rights reserved.

C:\Users\mchhetri>cd C:\Tools\JADE-all-4.5.0\JADE-bin-4.5.0\
C:\Tools\JADE-all-4.5.0\JADE-bin-4.5.0>dir
Volume in drive C is SWINBURNE
Volume Serial Number is 9A4F-48C4

Directory of C:\Tools\JADE-all-4.5.0\JADE-bin-4.5.0

01/08/2017  01:50 PM    <DIR>          .
01/08/2017  01:50 PM    <DIR>          ..
01/08/2017  01:50 PM    <DIR>          jade
               0 File(s)                0 bytes
               3 Dir(s) 23,955,869,696 bytes free

C:\Tools\JADE-all-4.5.0\JADE-bin-4.5.0>java -cp jade\lib\jade.jar jade.Boot -gui
Error: Could not find or load main class Jade.Boot

C:\Tools\JADE-all-4.5.0\JADE-bin-4.5.0>java -cp jade\lib\jade.jar jade.Boot -gui
Aug 01, 2017 1:57:32 PM jade.core.Runtime beginContainer
INFO: -----
      This is JADE 4.5.0 - revision 6825 of 23-05-2017 10:06:04
      downloaded in Open Source, under LGPL restrictions,
      at http://jade.tilab.com/
-----
Aug 01, 2017 1:57:33 PM jade.imtp.leap.LEAPIMTPManager initialize
INFO: Listening for intra-platform commands on address:
- jicp://136.186.6.197:1099

Aug 01, 2017 1:57:33 PM jade.core.BaseService init
INFO: Service jade.core.management.AgentManagement initialized
Aug 01, 2017 1:57:33 PM jade.core.BaseService init
INFO: Service jade.core.messaging.Messaging initialized
Aug 01, 2017 1:57:34 PM jade.core.BaseService init
INFO: Service jade.core.resource.ResourceManagement initialized
Aug 01, 2017 1:57:34 PM jade.core.BaseService init
INFO: Service jade.core.mobility.AgentMobility initialized
Aug 01, 2017 1:57:34 PM jade.core.BaseService init
INFO: Service jade.core.event.Notification initialized
Aug 01, 2017 1:57:34 PM jade.mtp.http.HTTPServer <init>
INFO: HTTP-MTP Using XML parser com.sun.org.apache.xerces.internal.jaxp.SAXParserImpl$JAXPS
Aug 01, 2017 1:57:34 PM jade.core.messaging.MessagingService boot
INFO: MTP addresses:
http://EN508D-MOHAM.ds.swin.edu.au:7778/acc
Aug 01, 2017 1:57:34 PM jade.core.AgentContainerImpl joinPlatform
INFO: -----
Agent container Main-Container@136.186.6.197 is ready.
-----
```

Figure 6 Steps to start JADE from command line

NOTE: please make sure that you use the correct naming convention – jade.Boot and not Jade.Boot (Java is case-sensitive). The –gui option is to tell JADE to start up the GUI.

NOTE: You need to tell Java (java.exe), which libraries to load (jade.jar) and which main class to load (jade.Boot) using the classpath option (-cp).

NOTE: In Java, you have to be familiar with the two environment variables - **PATH** and **CLASSPATH**. PATH is a variable, which provides a shortcut to all executables (e.g. java.exe). The CLASSPATH variables tells JAVA where to look for user defined classes (e.g. jade.jar).

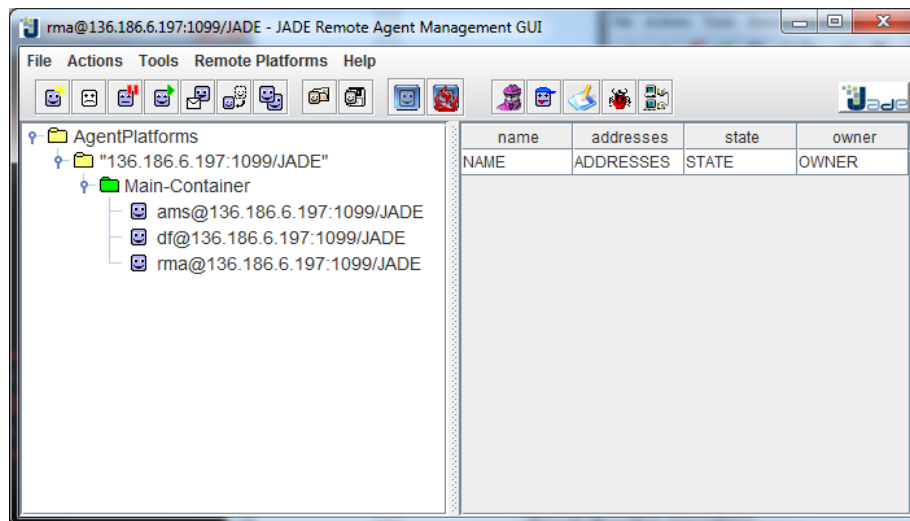


Figure 7 JADE Remote Agent Management GUI

By default, a Main-Container is created with three agents – ams, df and rma. If you can see the above GUI, you have installed JADE successfully.

JADE Examples

JADE comes with a number of example agents that are included in the **JADE-examples-4.5.0.zip** file. The **jadeExamples.jar** archive contains all the examples. The zipped folder also contains the source code for the examples.

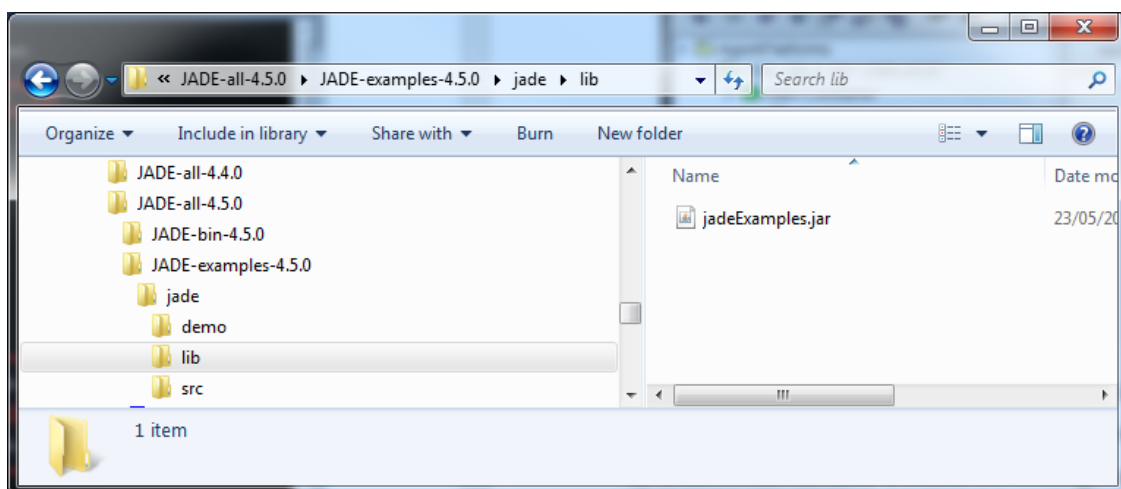
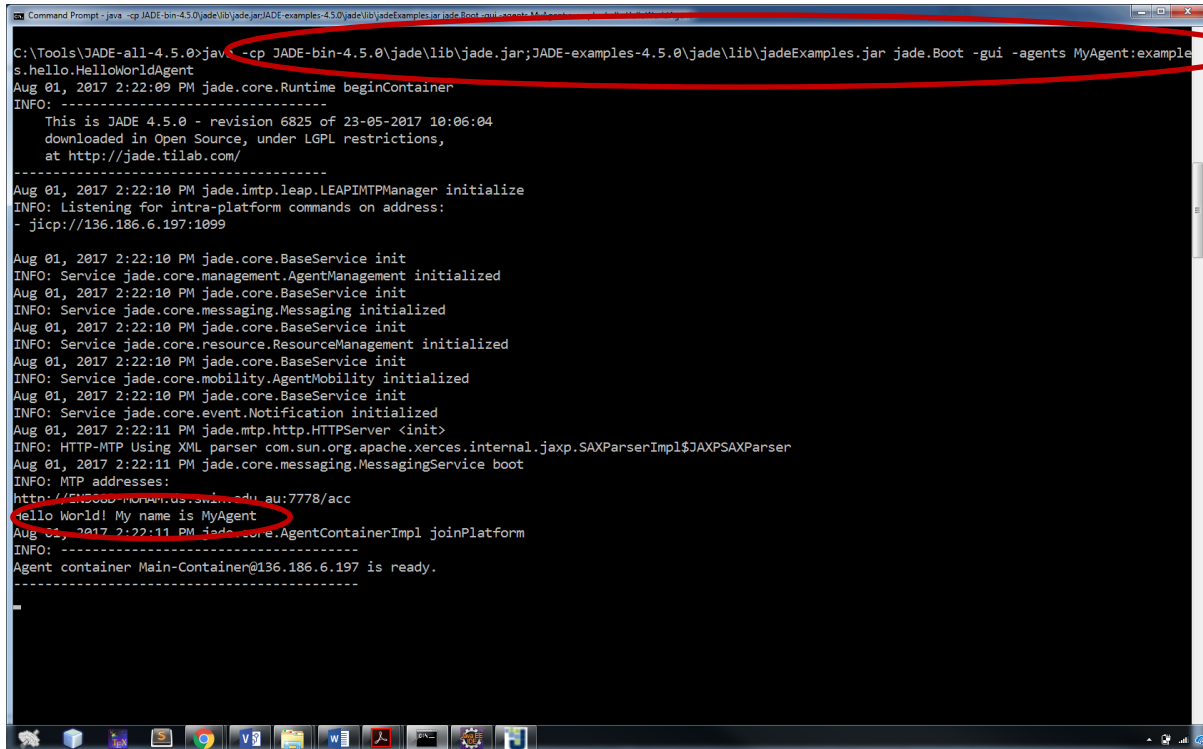


Figure 8 JADE examples archive file

Starting JADE with example agent

In order to start an example agent, we need to add jadeExample.jar to the CLASSPATH. In the screenshot below, we are creating a HelloWorldAgent (examples.hello.HelloWorldAgent) with the name MyAgent. On creation, the agent prints out “Hello world” and then terminates itself.



```
C:\Tools\JADE-all-4.5.0>java -cp JADE-bin-4.5.0\jade\lib\jade.jar;JADE-examples-4.5.0\jade\lib\jadeExamples.jar jade.Boot -gui -agents MyAgent:examples.hello.HelloWorldAgent
Aug 01, 2017 2:22:09 PM jade.core.Runtime beginContainer
INFO: -----
This is JADE 4.5.0 - revision 6825 of 23-05-2017 10:06:04
downloaded in Open Source, under LGPL restrictions,
at http://jade.tilab.com/
-----
Aug 01, 2017 2:22:10 PM jade.imtp.leap.LEAPIMTPManager initialize
INFO: Listening for intra-platform commands on address:
- jicp://136.186.6.197:1099

Aug 01, 2017 2:22:10 PM jade.core.BaseService init
INFO: Service jade.core.management.AgentManagement initialized
Aug 01, 2017 2:22:10 PM jade.core.BaseService init
INFO: Service jade.core.messaging.Messaging initialized
Aug 01, 2017 2:22:10 PM jade.core.BaseService init
INFO: Service jade.core.resource.ResourceManagement initialized
Aug 01, 2017 2:22:10 PM jade.core.BaseService init
INFO: Service jade.core.mobility.AgentMobility initialized
Aug 01, 2017 2:22:10 PM jade.core.BaseService init
INFO: Service jade.core.event.Notification initialized
Aug 01, 2017 2:22:11 PM jade.mtp.http.HTTPServer <init>
INFO: HTTP-MTP Using XML parser com.sun.org.apache.xerces.internal.jaxp.SAXParserImpl$JAXPSAXParser
Aug 01, 2017 2:22:11 PM jade.core.messaging.MessagingService boot
INFO: MTP addresses:
http://136.186.6.197:1099/acc
http://136.186.6.197:1099/acc
Hello World! My name is MyAgent
Aug 01, 2017 2:22:11 PM jade.core.AgentContainerImpl joinPlatform
INFO: -----
Agent container Main-Container@136.186.6.197 is ready.
-----
```

Figure 9 Creating HelloWorldAgent from command line

If you replace examples.hello.HelloWorldAgent with jade.core.Agent, it will create a standard JADE agent with the name MyAgent as shown below:

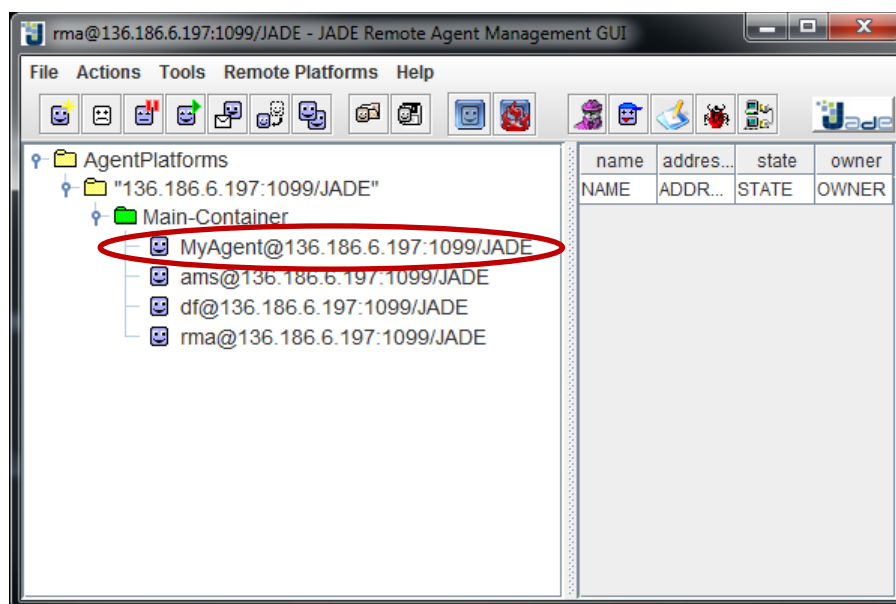


Figure 10 Creating standard JADE agent from command line

NOTE: Please terminate the JADE platform before restarting it since the platform can have only one main container.

Running JADE from inside Eclipse

In the previous examples, we ran JADE from the command line. However, we can do the same from inside an IDE such as Eclipse. The steps below show how to create a Java Project in Eclipse and run JADE from within the IDE.

Step 1: Configure Workspace. If you are using Eclipse for the first time, please get familiar with the concept of *workspace*. Eclipse uses the concept of *workspaces* to logically group related projects on your machine. When you start Eclipse, you can specify where you want to locate your workspace. By default, it creates a workspace folder within your home directory (see below). You can either use this or define your custom workspace.

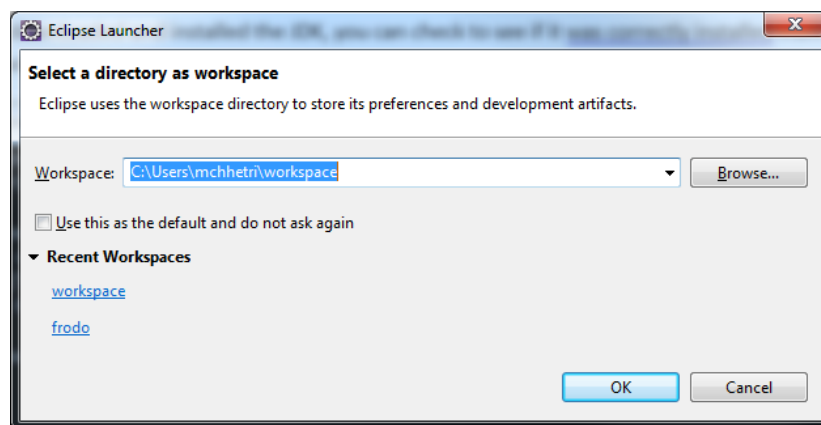


Figure 11 Configuring Eclipse workspace

Step 2: Create a Java Project. Since JADE is a java-based middleware, we create a JAVA project to run JADE from within the IDE. We use the wizard to do this – File → New → Java Project.

You can either use the default location, which is the workspace location or specify another location. Once you create the project, you can check the physical location of the project – C:\Users\mchhetri\workspace\COS30018Tutes (in this case).

Click next and finish creating the project.

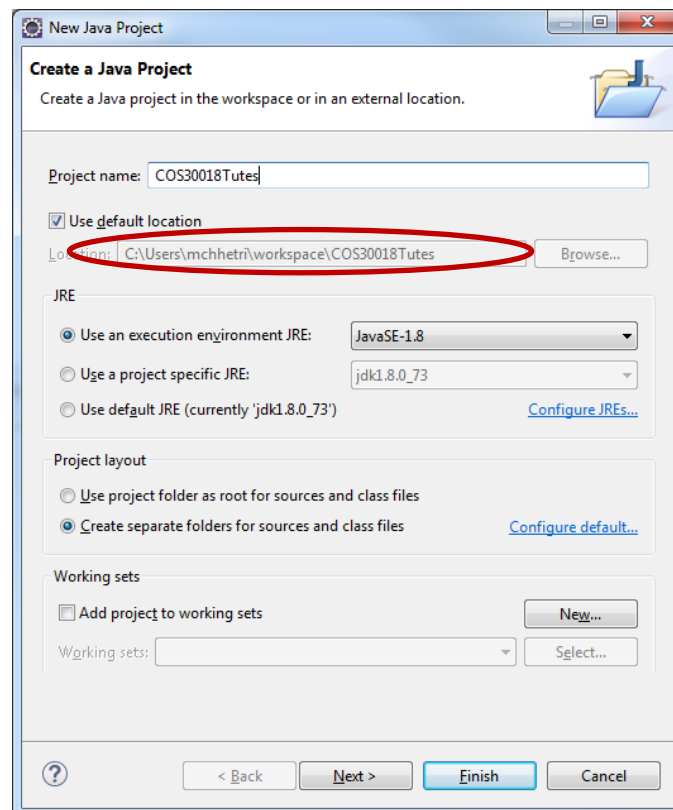


Figure 12 Creating a new Java Project called COS30018Tutes

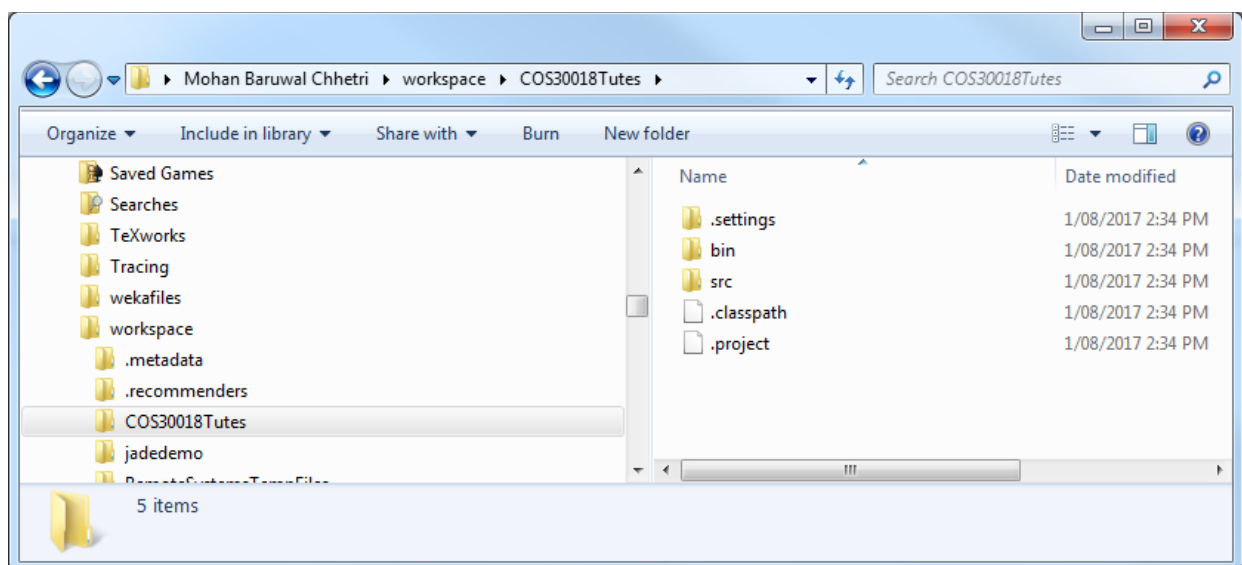


Figure 13 Physical location of project

By default, an Eclipse Java Project has two folders – **src**, which contains the source files (.java files) and **bin**, which contains the compiled files (.class files). To simplify the project setup, we can create a third folder called **lib**, where we will keep all our third-party jar files (e.g. jade.jar, jadeExamples.jar etc).

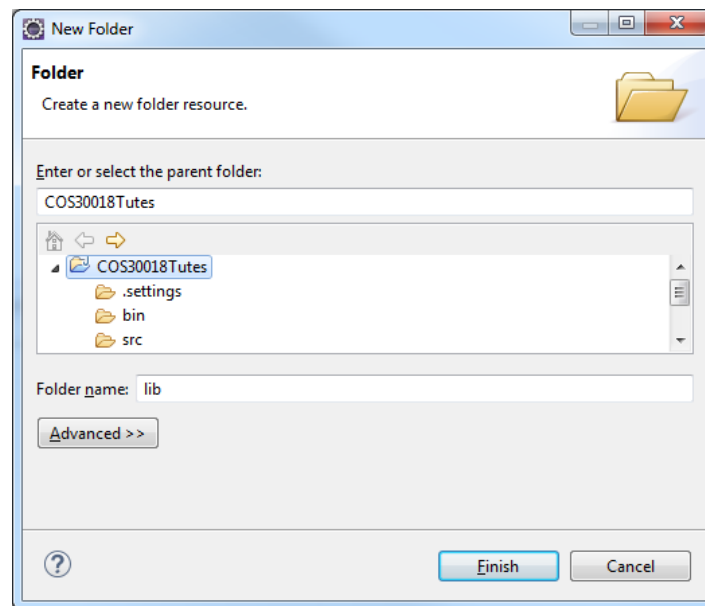


Figure 14 Creating a lib folder for the project

The updated project will look as shown below:

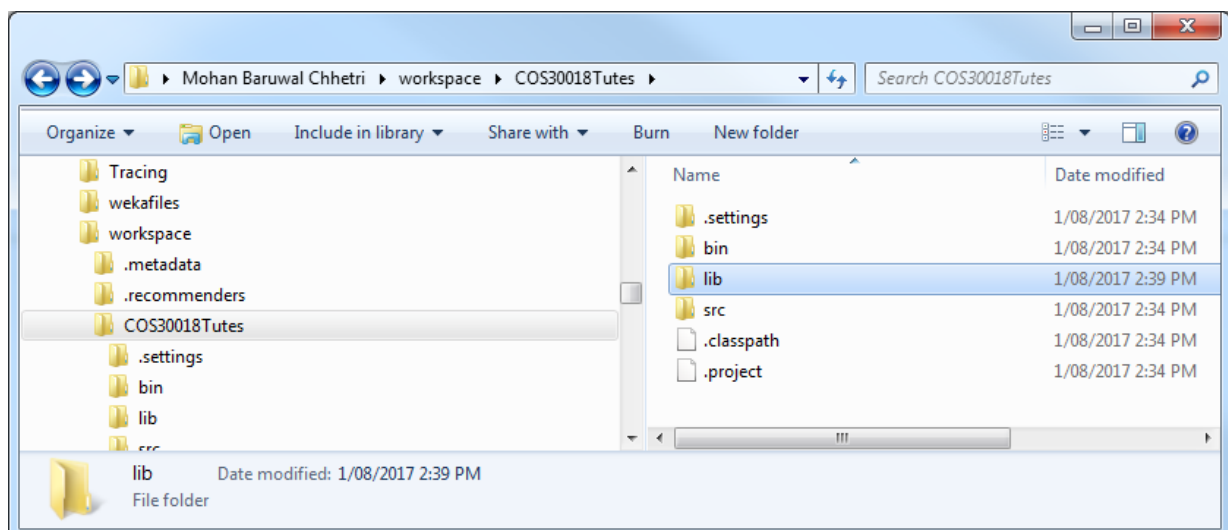


Figure 15 Project with src, bin and lib folders

Step 3: Add JADE libraries to project

Now, copy across the jade.jar and jadeExamples.jar files to the lib folder of your project. Either copy them directly to the physical location or inside the IDE and refresh.

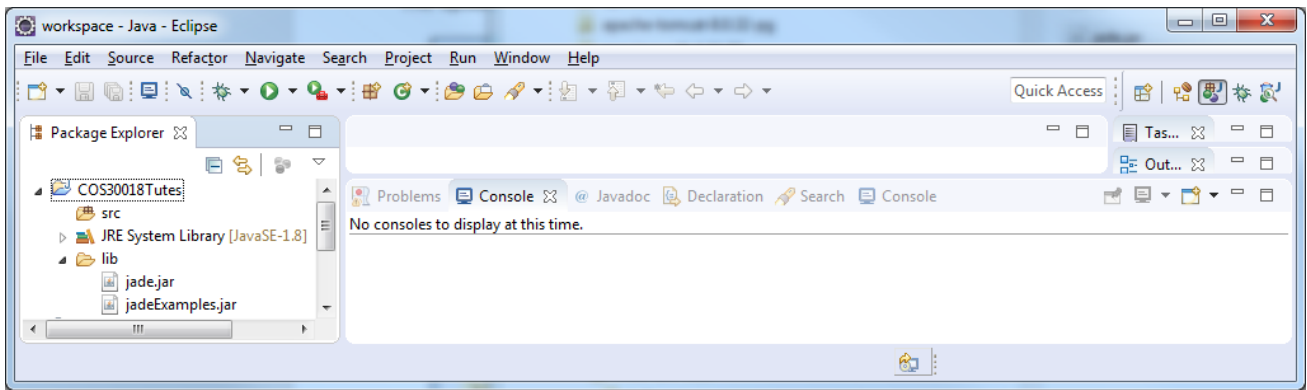


Figure 16 Adding Jar files to lib folder

Step 4: Update the project settings to specify the location of the JADE dependencies

Update the project settings and set the Java Build Path by adding the two jar files to it.

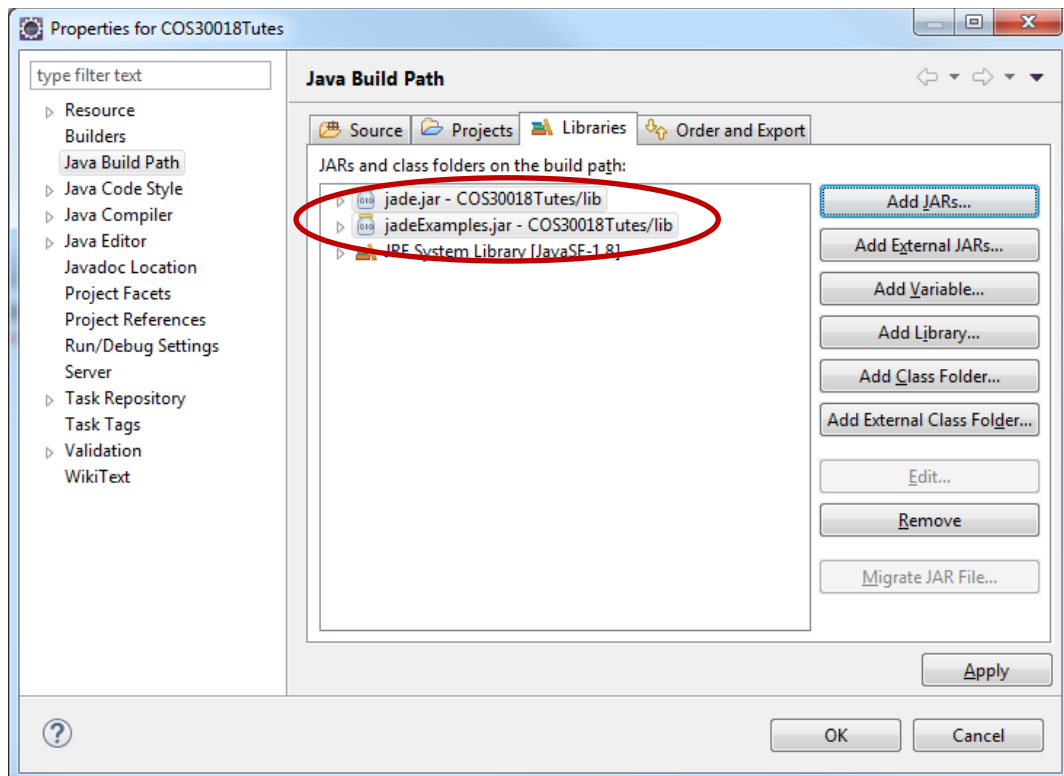


Figure 17 Updating Java Build Path

Step 5: Configure the project to run JADE from within the IDE

In Eclipse, we can define run configurations for our projects. Click on Run → Run Configurations. Select Java applications on the wizard and start a new configuration.

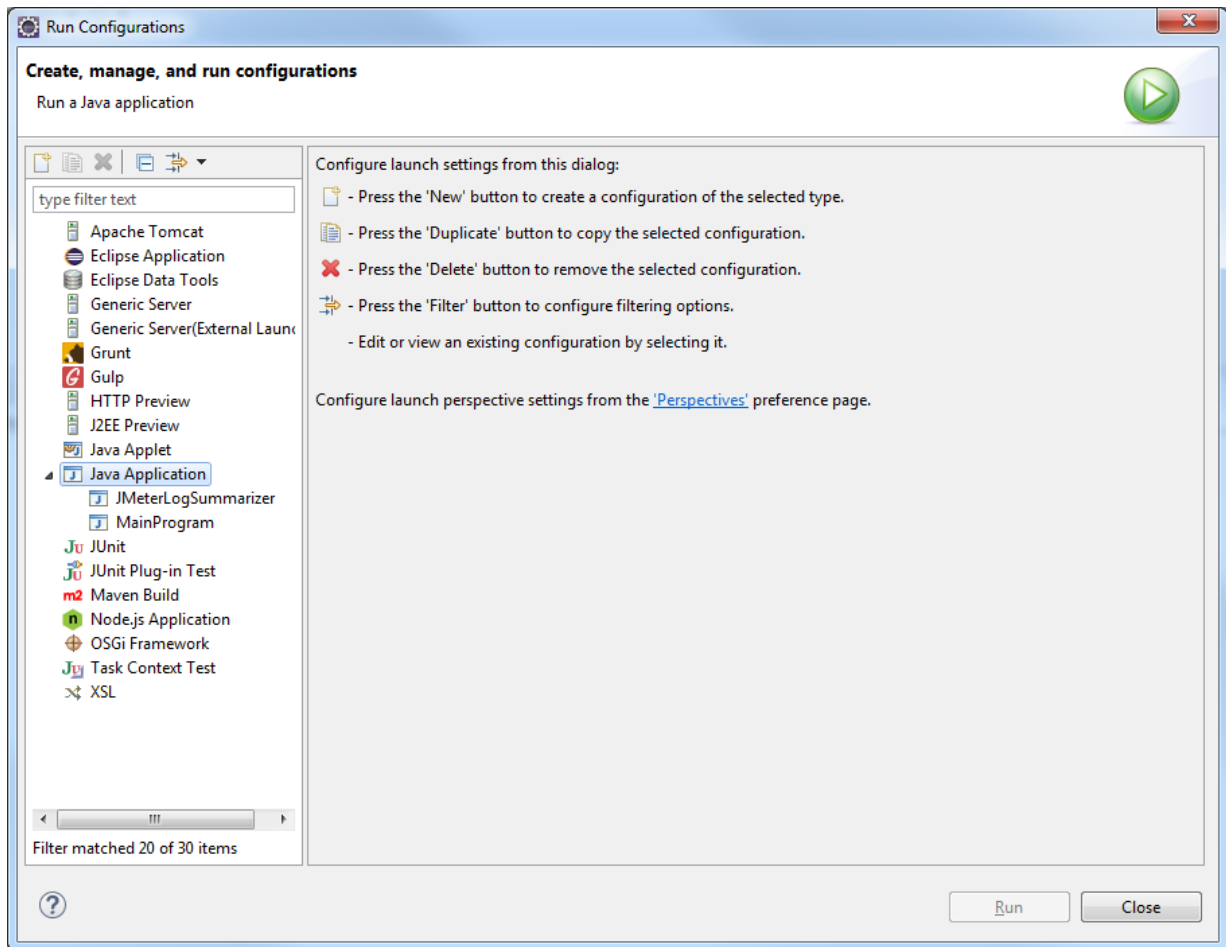


Figure 18 Select Java application and right-click to define new configuration

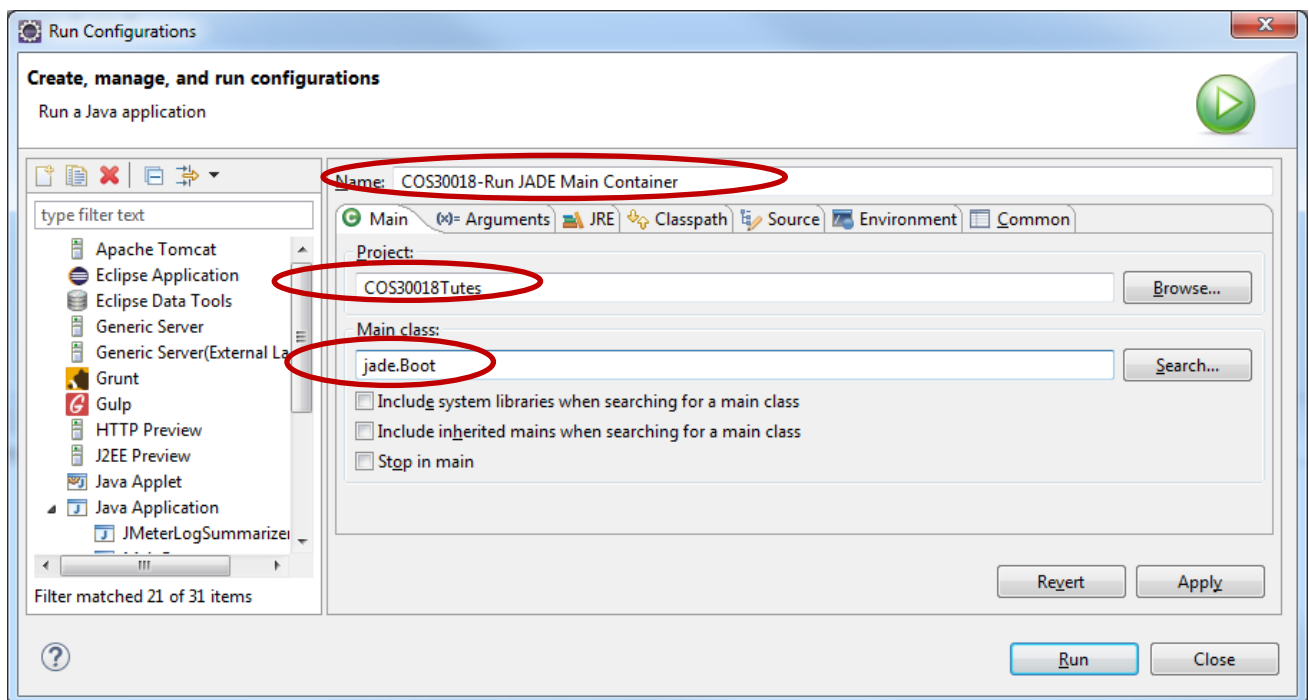
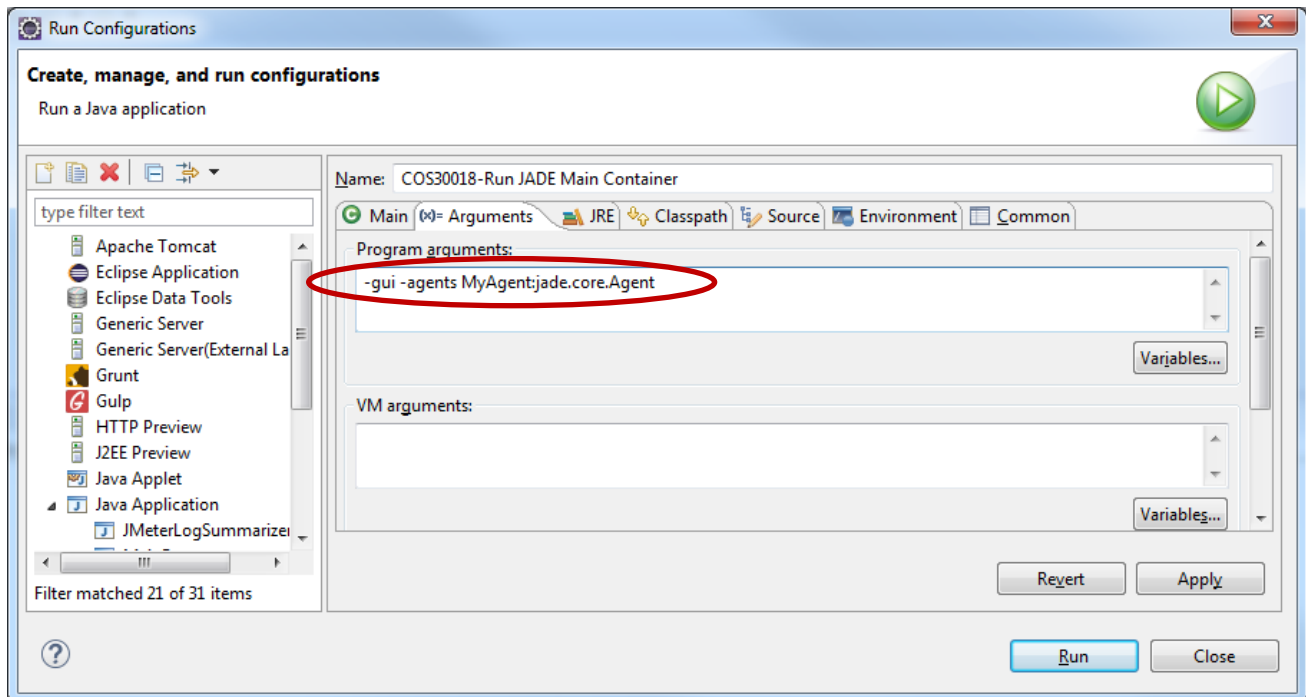


Figure 19 Specifying the run configuration name, the project to run within and the main class to run

In the above figure, we specify that we want to run the configuration within the previously created **COS30018Tutes** project. We name it **COS30018 – Run JADE Main Container** and specify the main class as **jade.Boot**. We also can specify any arguments that the Boot class takes (e.g. `-gui` to start the Graphical User interface). If we want to start up MyAgent, we can specify that as well using the `-agents` flag. The naming convention is `<AgentName>:<Agent Class>` i.e. `MyAgent:jade.core.Agent`



Apply the run configuration to save it. You can also run it to start up JADE. If you have typed in everything correctly, you should be able to see the JADE Remote Agent Management GUI and the MyAgent within the Main Container.

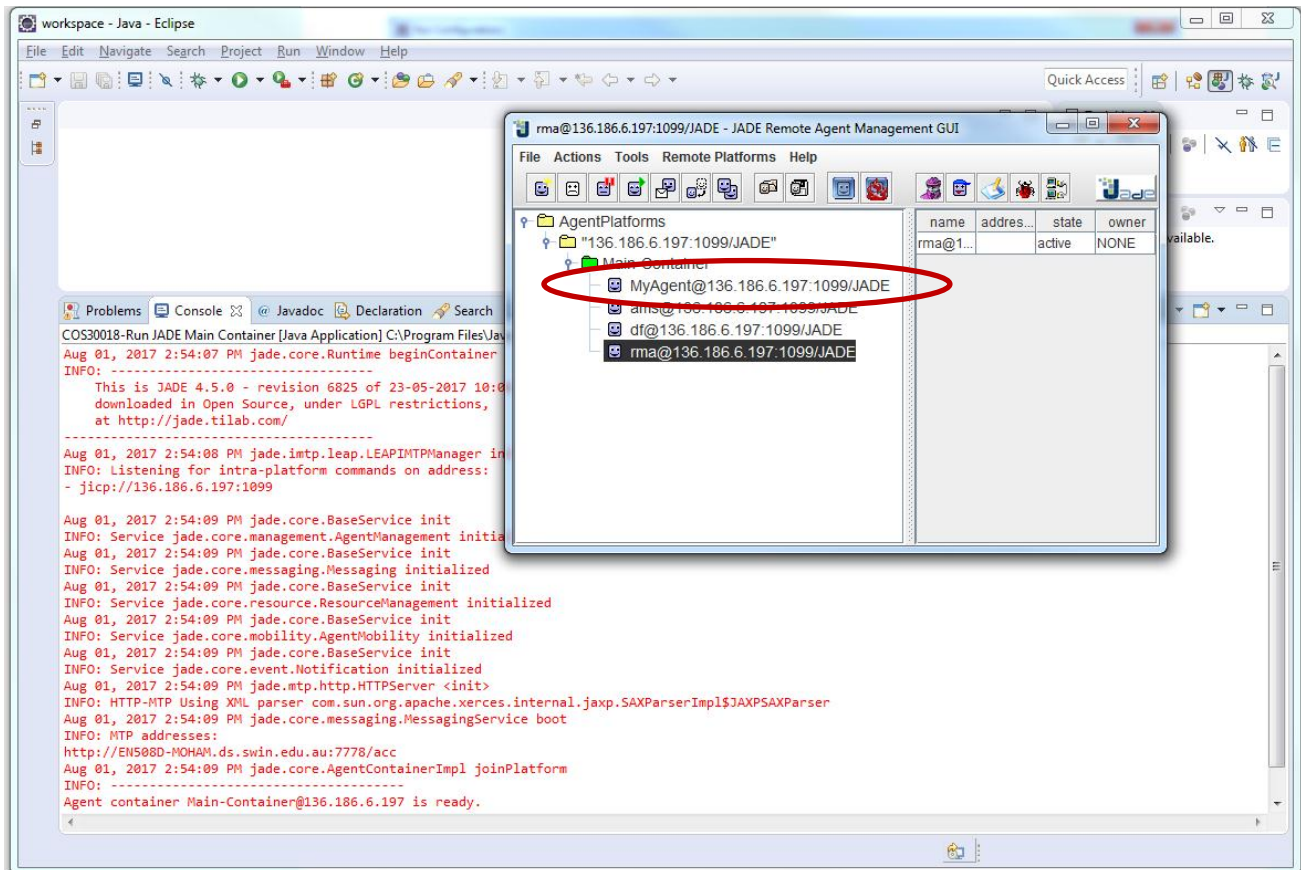


Figure 20 Starting up the JADE main container inside Eclipse

Congratulations! You have successfully configured your Eclipse IDE to launch the JADE platform within a self-contained JAVA project. Now you can start writing your own “intelligent agents”.